ALZ-PATHWAY-TWIN Sample

A true subset of the full dataset: 500 people, all 27 scenarios, all tables, hidden truth included. No registration.

500	27	13,500	23 MB
people drawn at random	scenarios, all kept	paired trajectories	compressed size

Specimen record, one person across worlds

subject_id	age_baseline	sex	apoe_e4_alleles
155	**57.75**	**female**	**1**
subtype_true	amyloid_onset_age	dementia age, S00	dementia age, S15
5	**50.2**	**59.75**	**62.75**

Open it, run your pipeline on it, then decide whether the full ten thousand are worth buying.

All data are synthetic and describe no real person. Effect shares in the scenario grid are hypotheses, not measured quantities, and nothing here is medical advice or a substitute for clinical evidence. The tables report where the model does not match published numbers, and the passport lists them.

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01 What the sample is

The sample is a real subset of the dataset, not a recomputed or simplified copy. It holds the first 500 person numbers of a random permutation of the 10,000 people (permutation seed 20261108). Every table is present, every one of the 27 scenarios is present for each of the 500 people, and the truth tables are included. A pipeline written against the sample runs unchanged against the full dataset, because the layout, column names and types are identical.

> **WHY IT IS A SUBSET AND NOT A SIMPLIFIED COPY**
> Because the people are the same people as in the full dataset, numbers you compute on the sample (an effect estimate, a forecast score) can be compared with the full-data result. Small-sample noise will differ, so expect wider uncertainty.

02 What you can and cannot do with it

- **Can:** test loading, joins on subject_id, paired-contrast code, an effect estimator against the hidden individual effect, a five-year forecast pipeline.
- **Cannot:** reliable power calculations or subtype-level conclusions. With 500 people, some outcomes are rare (dementia in 72 people in the untreated scenario) and subtype 3 has only a few dozen carriers of any event.
- **Cannot:** be used as training input for a model that is then scored on the full data: the 500 people are part of the full data.

03 Honest limitations

- **The product does not say which theory is true.** It shows what happens under each, and that the natural course is identical under all of them.
- **The link from biomarker to clinic is known only for anti-amyloid treatment at the symptomatic stage.** For clearance and inflammation the shares are a hypothesis axis whose ends rest on null trials and on the amyloid share.
- **The amyloid share (0.60) was derived from one trial of one drug.** The transfer to a second drug was tested against a published result and missed: predicted slowing 0.27 against 0.36 published.
- **No dementias of other kinds.** Vascular, frontotemporal, Lewy body and mixed forms are not modelled, so MCI-to-dementia conversion is faster than in population cohorts (model 0.19 per year against 0.049 published for population cohorts and 0.096 for specialist clinics).
- **Time from first symptoms to dementia is too long.** The model gives 4.5 years from CDR-SB 0.5 to dementia against 3.3 published; it could not be matched together with the 18-month decline rate.
- **Almost everyone reaches the MCI stage.** 96.8 % of the untreated cohort crosses CDR-SB 0.5 within 25 years, because the model has no separate boundary for normal ageing. Dementia (CDR-SB 4.5) is reached by 15.1 %.

- **Subtypes come from one CSF study.** Their shares and biology are taken from it; replication in other cohorts was not available. Subtype survival order was predicted and missed. The model gives weaker APOE e4 enrichment among dementia cases than the study (about 30 points lower).
- **Therapy matched to subtype in world W4 is an experimental condition,** derived from the biological description of the subtypes. No trial has tested it. The subtype assay is error-free.

The remaining limitations of the full dataset (no dropout, start at 50, no symptomatic drugs, design-value incidence, pre-symptomatic share not separable in a symptomatic trial) apply to the sample unchanged and are listed in the data passport.

PART II · TECHNICAL SPECIFICATION

04 Rows in the sample

Table	Rows in the sample
cohort	500
scenarios	27
visits	245,973
pet_scans	43,030
detection_events	4,678
treatments	4,678
adverse_events	148
subtype_assays	1,128
outcomes	13,500
truth: subject_truth	500
truth: theta_by_subject	13,500
truth: truth_quarterly	957,431

05 How the 500 people were chosen

Person numbers 1 to 10,000 were shuffled by a seeded pseudo-random permutation (seed 20261108); the first 500 of the permutation were kept and stored in ascending order. Each table was then filtered to those subject_id values. Nothing was recomputed, so every value is identical to the corresponding row of the full dataset. The sample therefore inherits the full dataset's age range (50 to 75), sex mix and APOE e4 mix up to sampling noise.

Cohort feature	Sample (500)	Full (10,000)
Mean age at entry	62.42	62.48
Female share	0.486	0.500
APOE e4 carriers (1 or 2 alleles)	0.206	0.226

06 Defects found and fixes applied

Defect found	Fix applied
A sample drawn as the first 500 numbers would inherit any ordering in the person numbering.	Numbers are drawn through a seeded permutation.
Filtering by subject_id alone would break partitioned tables that are stored per scenario.	Each scenario partition is filtered separately and keeps its path.

Defect found	Fix applied
Run metadata in the Parquet files describes the full run (n = 10,000).	Left unchanged on purpose: the sample is a subset of that run. Row counts differ from the metadata n.

07 Assumptions log

Assumption	Basis
A random 500 of 10,000 represents the cohort	Random permutation; compared in the table above
Pipelines carry over unchanged to the full data	Identical schema and column types
Hidden truth in the sample is safe to publish	All data are synthetic

08 Loading the files

Tables are Apache Parquet. Per-scenario tables are stored as one partition per scenario. A minimal session in Python:

```
import pandas as pd
cohort   = pd.read_parquet("cohort.parquet")
outcomes = pd.read_parquet("outcomes.parquet")
visits   = pd.read_parquet("visits/scenario_id=S15/part-0.parquet")
ctrl     = outcomes[outcomes.scenario_id == "S00"]
trt      = outcomes[outcomes.scenario_id == "S15"]
effect   = trt.set_index("subject_id").dementia_free_years \
         - ctrl.set_index("subject_id").dementia_free_years
```

The last three lines give the individual effect of screening plus lecanemab on dementia-free years for every person: the quantity real studies cannot observe.

09 Terms used

Term	Meaning
CDR-SB	Clinical Dementia Rating sum of boxes, 0 to 18; higher is worse. 0.5 marks the first symptoms (mild cognitive impairment stage here) and 4.5 the start of dementia.
MMSE	Mini-Mental State Examination, 0 to 30; higher is better.
p-tau217	Plasma concentration of tau phosphorylated at threonine 217, in pg/mL; two cut points define negative, intermediate and positive zones.
centiloid (CL)	Standard scale of amyloid PET signal; 0 is the young-control level and 100 the typical Alzheimer level. Positivity is set at 20 CL.
ARIA-E	Amyloid-related imaging abnormality with oedema, a side effect of anti-amyloid antibodies.
APOE e4	Gene variant that raises the risk of Alzheimer's disease; carried as 0, 1 or 2 alleles.
PEHE	Root mean squared error of the individual treatment effect estimate.